

Pulsed Nuclear Magnetic Resonance

PHYS3111 Quantum Physics

Student Notes

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1 Experimental Aim

To investigate the properties of nuclear spin relaxation in a number of materials and relate those measurements to the local magnetic environment within those materials. To detect “chemical shift” in materials and relate that to the structure of molecules.

1.1 WARNINGS!

- The permanent magnet in the PS2 Spectrometer sets up a magnetic field strong enough to damage magnetically encoded data, keep such objects away from the magnet.
- The magnet is very sensitive to the presence of nearby ferromagnetic materials which can significantly alter its field. Do not place any ferromagnetic objects, or electrical current conductors close to the magnet.
- The sample tubes are fragile, ensure not to drop them, and never force them into the spectrometer.

2 Background

Atomic nuclei are characterised by states which are inherently quantum mechanical in their behaviour, however, when performing nuclear magnetic resonance experiments we are dealing with vast multitudes of individual nuclei each acting largely independently. When dealing with such an ensemble, a classical description is usually sufficient and more easily understood, however the quantum mechanics is never far below the surface. The following theory will be kept in quantum mechanical terms for the most part, however at times classical terminology will be applied [2].

2.1 Basic Quantum Mechanics

Many nuclei, but not all, possess both a magnetic moment, μ , and an angular momentum, J . Such particles are said to have spin. When the angular momentum and magnetic moment are collinear, as they are in protons (hydrogen nuclei) and fluorine nuclei, both the particle and its surroundings can be investigated using various techniques of Nuclear Magnetic Resonance Spectroscopy. For these particles, the relation between μ and J is given by the vector equation:

$$\mu = \gamma J \quad (1)$$

the quantity γ , defined as the gyromagnetic ratio, is an important signature by which the particle can be identified and its behaviour predicted or interpreted.

The nuclear angular momentum is quantised in units of $\hbar$ as:

$$J = \hbar \hat{s} \quad (2)$$

where $\hat{s}$ is the vector operator of the spin of the nucleus. We often discuss "spin states", which may be described in terms of a set of discrete values for the component of the angular momentum m , measured along some axis, z (chosen as it will be the direction of the static magnetic field). m can take integer or half integer values in the range:

$$m = -s, -s + 1, \dots, s - 1, s \quad (3)$$

where s is the intrinsic angular momentum quantum number (sometimes called the spin quantum number). For the hydrogen (1H) and fluorine (^{19}F) nuclei, which we are concerned with in this experiment, $s = \frac{1}{2}$. Therefore m can take two possible values, $m = \pm \frac{1}{2}$, which can be represented as column vectors as follows:

$$|+\frac{1}{2}\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |-\frac{1}{2}\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (4)$$

Hence the wavefunction for our nucleus is:

$$|\psi\rangle = a_{+\frac{1}{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_{-\frac{1}{2}} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (5)$$

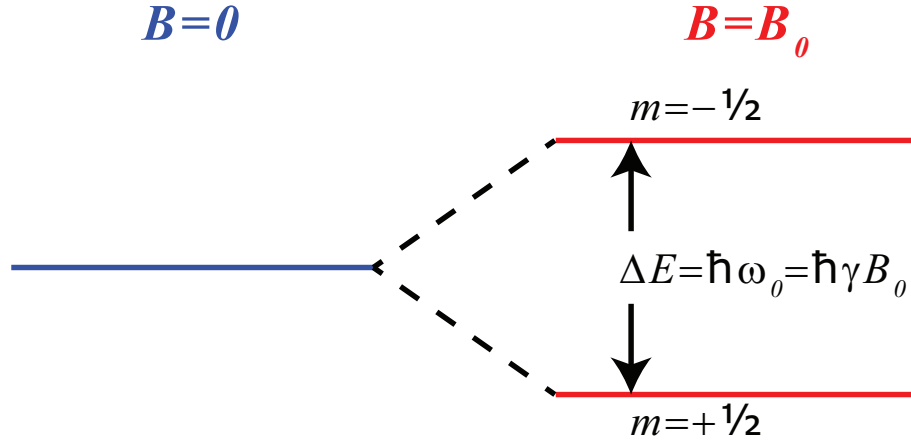


Figure 1: Energy state separation in a magnetic field

where $a_{+\frac{1}{2}}$ and $a_{-\frac{1}{2}}$ are complex amplitudes. And our spin operator can be defined using the Pauli matrices, $\hat{\mathbf{s}} = \frac{1}{2}\hat{\boldsymbol{\sigma}}$, with:

$$\hat{\sigma}_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \hat{\sigma}_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \hat{\sigma}_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (6)$$

When we introduce the nucleus to an external magnetic field $\mathbf{B}$ in the z -direction, the magnetic moment interacts with the field as described by the following Hamiltonian:

$$\hat{H} = -\boldsymbol{\mu} \cdot \mathbf{B} = -\mu_z B_0 = -\gamma \hbar \hat{s}_z B_0 = -\frac{\gamma \hbar B_0}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (7)$$

This magnetic interaction is known as the Zeeman interaction, and it yields the energy level structure shown in Figure 1, with the energy splitting between the two spin states being:

$$\Delta E = \gamma \hbar B_0 \equiv \hbar \omega_0 \quad (8)$$

ω_0 is known as the Larmor frequency, and it will be explained shortly.

The time-dependent Schrödinger equation of the system is given by:

$$i\hbar \frac{\delta}{\delta t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle \quad (9)$$

If $\hat{H}$ is constant with time, then this yields:

$$|\psi(t)\rangle = \exp\left(-\frac{i\hat{H}t}{\hbar}\right) |\psi(0)\rangle = \exp(i\gamma B_0 \hat{s}_z t) |\psi(0)\rangle \equiv U(t) |\psi(0)\rangle \quad (10)$$

$U(t)$ is known as the evolution operator, and its form, $U(t) = \exp(i\gamma B_0 \hat{s}_z t)$ is identical to a clockwise rotation of the state about the z -axis by an angle $\gamma B_0 t$, hence it can be said that the spin precesses around the magnetic field direction at the Larmor frequency ω_0 defined above.

We can now start to think of an ensemble of nuclei rather than a single nucleus in isolation. The nuclei in the ensemble may occupy many different states as such we can deal in an average expectation value. The z component of the angular momentum is hence given by:

$$\overline{\langle \psi | \hat{s}_z | \psi \rangle} = \frac{1}{2} \left(\overline{|a_{+\frac{1}{2}}|^2} - \overline{|a_{-\frac{1}{2}}|^2} \right) \quad (11)$$

This equation can be interpreted as the population difference in the ensemble between the $m = +\frac{1}{2}$ and $m = -\frac{1}{2}$ spin states, i.e. the ensemble averaged expectation value of $\hat{s}_z$ is the spin polarisation of the ensemble.

In this experiment the component of magnetisation perpendicular to the magnetic field is what we actually measure, as such it is interesting to perform the same ensemble averaged expectation value operation on $\hat{s}_x$ to yield:

$$\overline{\langle \psi | \hat{s}_x | \psi \rangle} = \frac{1}{2} \left(\overline{a_{+\frac{1}{2}}^* a_{-\frac{1}{2}}} + \overline{a_{+\frac{1}{2}} a_{-\frac{1}{2}}^*} \right) \quad (12)$$

which represents the phase coherence between the $m = +\frac{1}{2}$ and $m = -\frac{1}{2}$ states.

In semi-classical terms, eqn (11) represents the longitudinal component of the net magnetisation, in thermal equilibrium there are more nuclei aligned parallel to the magnetic field than anti-parallel and so the net magnetisation in z is given by $M_z = \mu (N_{parallel} - N_{anti-parallel})$. Similarly, equation eqn (12) represents the transverse component of the net magnetisation, in equilibrium there is no phase coherence in the spin precession, as such the transverse magnetisation is zero.

2.2 Resonant Excitation

At this stage we have shown that in a magnetic field the spin precesses around the magnetic field vector with a frequency ω_0 . Further, for an ensemble of spins at equilibrium, fewer spins are precessing about $-\mathbf{B}_0$ (in the $m = -\frac{1}{2}$ state) than about $+\mathbf{B}_0$ (in the $m = +\frac{1}{2}$ state), as such there is a net positive angular momentum, and as there is no phase coherence between spins at equilibrium, all components perpendicular to the magnetic field average to zero in the ensemble.

We now introduce an oscillating magnetic field applied perpendicular (arbitrarily chosen as the x -direction) to our static magnetic field $\mathbf{B}_0$, this is our linearly polarised RF pulse $2B_1 \cos(\omega t)$. This modifies the Hamiltonian in the laboratory frame as follows:

$$\hat{H}_{lab} = -\gamma \hbar B_0 \hat{s}_z - 2\gamma \hbar B_1 \cos(\omega t) \hat{s}_x \quad (13)$$

The linearly polarised oscillating $2B_1 \cos(\omega t)$ can be decomposed into a pair of circularly polarised fields, each of amplitude B_1 , rotating in opposite directions in the x - y plane as shown in Figure 2. One of these components will be rotating in the same direction as the spin precession, it is this component which will be responsible for the phenomenon of nuclear magnetic resonance, the component rotating counter to the spin precession can be neglected as $B_0 \gg B_1$.

We can next introduce a transformation into frame rotating with the spin precession, in which the field B_1 is stationary. In this frame, the Hamiltonian is time independent, and

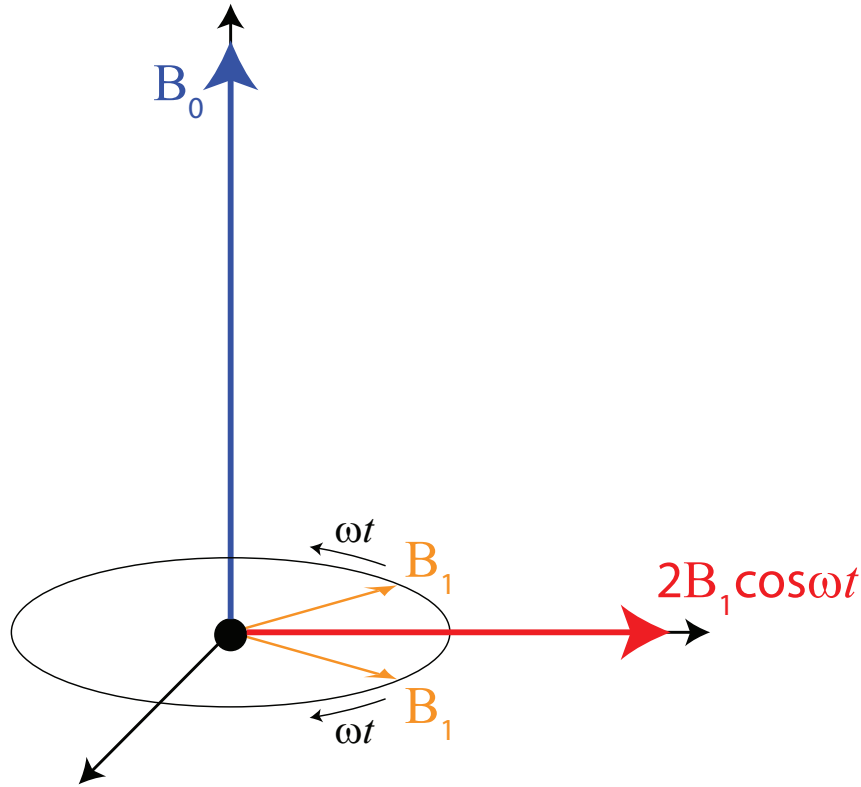


Figure 2: Linearly polarised, oscillatory transverse field $2B_1 \cos(\omega t)$, represented as a pair of counter-rotating circularly polarised fields of magnitude B_1 .

is as follows:

$$\hat{H}_{rot} = -\boldsymbol{\mu} \cdot \mathbf{B}_{eff} = -\gamma\hbar \left(B_0 - \frac{\omega}{\gamma} \right) \hat{s}_z - \gamma\hbar B_1 \hat{s}_x^* \quad (14)$$

$$\mathbf{B}_{eff} = \left(B_0 + \frac{\omega}{\gamma} \right) \hat{\mathbf{z}} + B_1 \hat{\mathbf{x}}^* \quad (15)$$

Suppose now that we tune the frequency of B_1 to the Larmor frequency ($\omega = \omega_0$), this is the resonance condition, such that:

$$\frac{\omega}{\gamma} = \frac{\omega_0}{\gamma} = B_0 \quad (16)$$

$$\Rightarrow \mathbf{B}_{eff} = B_1 \hat{\mathbf{x}}^* \quad (17)$$

In this case $\mathbf{B}_{eff}$ is a constant magnetic field in the x^* direction, and hence the spins begins to precess about $\mathbf{B}_1$ at a rate $\Omega = \gamma B_1$ (in the rotating frame). In the lab frame the spins are still precessing about B_0 , but an additional torque has been applied rotating them towards the x - y plane. If we apply the oscillating field for just the right amount of time, the spins will rotate by 90° and will be in non-thermal equilibrium rotating, in phase, about $\hat{\mathbf{z}}$ in the x - y plane. This is called a 90° (or $\frac{\pi}{2}$) pulse.

If we apply the oscillating field for twice this amount of time, the spin will rotate by 180° and will precess about $-\hat{\mathbf{z}}$. This is called a 180° (or π) pulse.

2.3 Relaxation

The effect of a resonant RF pulse is to disturb the spin system from its thermal equilibrium state, however, over time the system will return to equilibrium. As discussed above, at equilibrium there are two conditions which must be met:

There are more nuclei aligned parallel to the magnetic field than anti-parallel and so the net magnetisation in z at equilibrium is $M_z = M_0 = \mu(N_{parallel} - N_{anti-parallel})$. The system returns to M_0 via a process known as spin-lattice relaxation, wherein energy from the surroundings (referred to as the lattice) is transferred to the spin system. This process is described by:

$$\frac{dM_z}{dt} = -\frac{(M_z - M_0)}{T_1} \quad (18)$$

$$\Rightarrow M_z(t) = M_z(0) \exp\left(-\frac{t}{T_1}\right) + M_0 \left(1 - \exp\left(-\frac{t}{T_1}\right)\right) \quad (19)$$

Where the characteristic time of the relaxation T_1 , is known as the spin-lattice, or longitudinal, relaxation time.

We should emphasise that using this setup we are only able to detect transverse magnetisation $M_{x,y}$, as such we have no way of directly determining M_z and we must employ a more sophisticated method, known as inversion-recovery, to determine T_1 .

The second condition of thermal equilibrium is that there is no transverse magnetisation ($M_{x,y} = 0$), relaxation to this state is governed by a process called spin-spin relaxation where the spins come to thermal equilibrium among themselves, i.e. the spins undergo phase decoherence. This process is described by:

$$\frac{dM_{x,y}}{dt} = -\frac{M_{x,y}}{T_2} \quad (20)$$

$$\Rightarrow M_{x,y}(t) = M_{x,y}(0) \exp\left(-\frac{t}{T_2}\right) \quad (21)$$

Where the characteristic time of the relaxation T_2 , is known as the spin-spin, or transverse, relaxation time.

Although we can measure $M_{x,y}$ directly in this experiment, in practice the inhomogeneity in the magnetic field across the sample volume washes out the effect of spin-spin relaxation in a simple measurement. This means that upon application of a 90° pulse (rotating the magnetisation into the x - y plane) the varying magnetic field felt by spins at different points in the sample contributes more to spin de-phasing than does spin-spin interactions. As such we will use an effect known as the Hahn echo to determine T_2 .

3 Prework Questions

Question 1 Given that the Larmor frequency of f_0 , of protons, in our spectrometer is ~ 21.7 MHz and the gyromagnetic ratio γ is $2.675 \times 10^8 \text{ rad}\cdot\text{s}^{-1}\cdot\text{T}^{-1}$, what is the

magnitude of the magnetic field B_0 set up by the permanent magnet, at the sample position?

Hint: Refer to eqn. 8 and the fact that $\omega_0 = 2\pi f_0$

Question 2 What do you expect the Larmor frequency of ^{19}F nuclei to be in the field from **Question 1**?

Question 3 If the Larmor frequency of hydrogen nuclei in a particular sample is 21.696 MHz, but the PS2 Synthesizer is set to 21.700 MHz, what would be the oscillation period of the beats appearing on Ch. 2 of the oscilloscope?

Question 4 When looking at chemical shift, the conventional parlance is refer to the shift in parts per million (ppm). If you found two peaks on you NMR spectrum (i.e. the Fourier transform of the output of Ch. 2 at a frequency away from the Larmor frequency), one at 20.0000 MHz and the other at 20.0001 MHz, what would be the chemical shift difference, expressed in ppm, between these peaks?

4 Experimental Instructions

Before proceeding, begin by following the setup procedures in the Operating Instructions.

After following these procedures you should have a decaying signal, the Free Induction Decay (FID) signal, from light oil on both channel 1 and 2 of the oscilloscope. You should also have the LabVIEW software open and running.

4.1 Experiment 1: FID Amplitude vs Time

This measurement is used to find a time constant known as T_2^* , which is the decay of transverse magnetisation due to inhomogeneity in the magnetic field over the sample volume. This inhomogeneity is reduced during setup by adjusting the magnetic field gradients, but it can never be completely removed.

Set the following parameters:

Setting	Value
Sample	L-OIL
A_len	$\sim 2.5 \mu\text{s}$
B_len	$0.02 \mu\text{s}$
Num_B	0
tau	0.0050 s
Period	300.0 ms
Sync	A
Pulse A	On
Pulse B	Off

You may need to adjust A_len slightly to achieve the maximum amplitude of the FID, this maximum corresponds to applying a 90° pulse.

Acquire the FID and find T_2^* using the **T2*** tab of LabVIEW program.

4.2 Experiment 2: FID Amplitude vs RF Pulse Length

Now we will explore the relationship between FID Amplitude vs RF Pulse Length more thoroughly and determine the times for a 90° and 180° pulse time to be used in later experiments.

Set the following parameters:

Setting	Value
Sample	L-OIL
B_len	$0.02\ \mu\text{s}$
Num_B	0
tau	$0.0050\ \text{s}$
Period	$300.0\ \text{ms}$
Sync	A
Pulse A	On
Pulse B	Off

Using the **FID Amplitude** tab acquire the FID amplitude at a range of A pulse times from $0.02\ \mu\text{s}$ to $10.0\ \mu\text{s}$ at $0.50\ \mu\text{s}$ intervals. Be sure to not the sign of the signal on channel 2. Take extra data points in the region where the FID amplitude becomes zero ($\sim 5\ \mu\text{s}$).

Remember that the signals displayed on the oscilloscope are an indication of the magnitude of the transverse magnetisation, hence the length of pulse required to make the FID amplitude become 0 corresponds to a 180° flip in the magnetisation, record this 180° pulse time. Also divide the 180° pulse time by 2 to determine your 90° pulse time.

Question 5 Why is it not best practice to determine the 90° pulse length first and double it to get the 180° pulse length?

4.3 Experiment 3: T_2 Measurement (Two-Pulse)

This experiment uses a spin echo technique to remove the effects of magnetic field inhomogeneity which prevented us from measuring T_2 using the FID.

Set the following parameters to observe a spin echo:

Setting	Value
Sample	L-OIL
A_len	90° pulse length
B_len	180° pulse length
Num_B	1
tau	$0.0050\ \text{s}$
Period	$300.0\ \text{ms}$
Sync	A
Pulse A	On
Pulse B	On

In this technique, invented by Irwin Hahn in 1950, we first apply a 90° pulse, then after a delay τ , we apply a 180° pulse. After our initial 90° pulse, the spins in areas of

stronger than average fields precess faster than average and those in weaker fields more slowly. The spins "dephase" and the induced emf fades. After the 180° pulse, however, the spins that were "ahead" because they are in a stronger field are now "behind". Because their protons are precessing faster, however, they will now "catch up" to the "average". In the same way, after the 180° pulse, "slow" spins are now "ahead" and the "average" will overtake them. The spins rephase momentarily and dephase again. This is why the oscilloscope trace after the first 180° pulse shows a rise to a maximum and then a decay. The difference between the height of the initial signal and the echo maximum is due to the actual stochastic processes of T_2 .

Using the **T2** tab acquire the spin echo amplitude at a range of τ (tau) times from 5 to 70 ms at 5 ms intervals. Plot and fit the τ vs echo amplitude data to determine a value for T_2 .

4.4 Experiment 4: T_2 Measurement (CPMG Sequence)

The method used to determine T_2 in experiment 3, using 2 pulses and varying the time between them, is accurate for liquids where the self diffusion time of the spins through the magnetic field gradients is slow compared to T_2 . This is not the case for all liquids, so the two pulse method is flawed, additionally the method is rather time consuming. Both these issues can be circumvented by using a multiple pulse sequence.

The pulse sequence used in this method, devised by Carr and Purcell, consists of a 90° pulse, followed at time τ by a series of N 180° pulses, each separated by 2τ .

$$90^\circ - \tau - 180^\circ - 2\tau - 180^\circ - 2\tau - 180^\circ - 2\tau - \text{etc.}$$

This creates a series of echoes equally spaced between the 180° pulses with exponentially decaying amplitude. This exponential decay can be used to determine T_2 without the influence of diffusion.

Set the following parameters to observe this sequence:

Setting	Value
Sample	L-OIL
A_len	90° pulse length
B_len	180° pulse length
Num_B	30
tau	0.0050 s
Period	1.0 s
Sync	A
Pulse A	On
Pulse B	On
MG	Off

Acquire the decaying echo sequence using the **T2 CPMG** tab and determine T_2 from the decaying echo amplitude.

There is a practical problem with the Carr-Purcell sequence described above, if the pulse time is slightly off, the error is compounded by the end of the sequence. E.g. if the 180° pulse was actually rotating the magnetisation by 182° , by the 20th pulse the spectrometer

will have accumulated a rotational error of 60° . This rotational error yields values for T_2 which are too small.

Meiboom and Gill devised a sequence which would avoid this accumulated rotational error by adding a phase shift of 90° between the 90° and the 180° pulses. Repeat the above T_2 measurement with the **MG** switch in the On position.

4.5 Experiment 5: T_1 Measurement

Our apparatus is not sensitive to magnetisation in the z direction, as such when we apply a 180° pulse to flip the spin precession by 180° from $\hat{z}$ to $-\hat{z}$, we are unable to track the evolution of the magnetisation as it trends back towards equilibrium.

In order to track this evolution we instead apply a sequence known as inversion-recovery, wherein 180° pulse, then wait a time τ , before applying a 90° pulse. After the 180° pulse there is no transverse magnetisation, however the instantaneous longitudinal magnetisation M_z determines the amplitude of the FID after the 90° pulse. Hence by varying the time τ between pulses we can infer the longitudinal magnetisation over time.

Set the following parameters to observe the degree of inversion recovery at time τ :

Setting	Value
Sample	L-OIL
A_len	180° pulse length
B_len	90° pulse length
Num_B	1
tau	0.0100 s
Period	1.0 s
Sync	B
Pulse A	On
Pulse B	On

Using the **T1** tab acquire the inversion-recovery amplitude at a range of τ (tau) times from 10 to 300 ms at $\sim 5-10$ ms intervals. Plot and fit the τ vs FID amplitude data to determine a value for T_1 .

4.6 Experiment 6: The Chemical Shift

If we adjust the synthesiser frequency away from the Larmor frequency such that $\omega \neq \omega_0$ we observe a decaying, oscillatory signal on channel 2 of the oscilloscope. The frequency of this oscillation is the beat frequency between the synthesiser frequency and the Larmor frequency, however if nuclear spins in the sample exist in multiple different local magnetic fields, there will be multiple Larmor frequencies present in the sample. Hence by taking the Fourier transform of the oscillating FID signal we can identify slight shifts in magnetic field experienced by spins in the sample. These shifts are caused by varying chemical shielding of nuclear spins at different locations in the molecule.

Set the following parameters:

Setting	Value
Sample	L-OIL
A_len	90° pulse length
B_len	0.02 μ s
Num_B	0
tau	0.0050 s
Period	300.0 ms
Sync	A
Pulse A	On
Pulse B	Off

Tune the synthesiser frequency such that you see oscillations in the signal on channel 2 (adjust from the Larmor frequency by ~ 0.00005 MHz) and acquire the waveform on the **Chemical Shift** tab. The program will acquire waveforms for the specified number of averages then perform a Fast Fourier Transform. For light oil we expect only a single peak as the hydrogen nuclei sites in light oil are all roughly equivalent.

Perform the same process with the ethanol sample instead (some adjustment of the synthesiser frequency and number of averages may be required for best results), in order to show that ethanol has three peaks, hence three inequivalent sites of hydrogen nuclei.

For all chemical shift measurements it may be useful to first display the FFT of channel 2 on the oscilloscope directly, do this by using the **FFT** button on the oscilloscope.

4.7 Experiment 7: T_1 Measurement in Heavy Oil

Repeat the measurement in experiment 5 in the heavy oil (H-OIL) sample.

Note: With the heavier (thicker/more viscous) oil you will need to modify the size and range of τ steps covered.

4.8 Optional Experiments

These experiments are optional extensions and have not had the same degree of development as the previous experiments. In some cases there is not software tailored to the data acquisition, so you'll likely have to manually acquire data using the Measure/Cursor functions on the oscilloscope. In all cases the onus is on you to determine how you would like to run the experiment in order to demonstrate what you have learned throughout the experiment.

Additional experiments are also described in the Instructor's Manual[1], however the required samples may not be available.

4.8.1 pNMR in Teflon

Using the solid sample of Teflon measure the value of T_2^* , T_2 and T_1 of the fluorine nuclei in Teflon.

Hint: First calculate the resonance frequency of ^{19}F nuclei in our magnetic field.

4.8.2 Chemical Shift in FC-70 Fluoroinert

Following the method of experiment 6, measure the chemical shift at different ^{19}F nuclei molecular positions in a sample of FC-70.

4.8.3 T_1 in tap water and in a dilute Fe^{3+} solution

Prepare a sample of tap water and determine T_1 using the inversion recovery method.

Hint: T_1 is very long in tap water use a pulse repetition time of ~ 10 s.

Compare this to that found in a dilute Fe^{3+} solution.

Glossary

180° Pulse a pulse of oscillating magnetic field, at the Larmor frequency, of a duration just sufficient to rotate the magnetisation by 180°, sometimes called a π pulse.

90° Pulse a pulse of oscillating magnetic field, at the Larmor frequency, of a duration just sufficient to rotate the magnetisation by 90°, sometimes called a $\frac{\pi}{2}$ pulse.

Chemical Shielding the partial shielding of a nucleus, from the static applied magnetic field, by surrounding electrons.

Chemical Shift the difference in chemical shielding, in various parts of a molecule, resulting in different Larmor frequencies for nuclei in different molecular positions.

Fourier Transform transformation of the description of a function from one space (e.g. time) to the description via variables in the reciprocal space (frequency).

Free Induction Decay (FID) the decaying, oscillating (at the Larmor frequency) signal which follows application of a 90° pulse to nuclei.

Gyromagnetic Ratio the ratio of the nuclear magnetic moment of a nucleus to the spin angular momentum of the nucleus.

Inhomogeneity unevenness with respect to change in spatial position.

Inversion Recovery the recovery of the nuclear z -magnetisation (M_z), with time, after inversion by a 180° pulse.

Larmor Precession the precession of a nuclear magnetic moment about a steady magnetic field.

Magnetic Field Gradient the variation of a magnetic field as a function of spatial position.

Magnetic Resonance Imaging (MRI) an imaging technique employing NMR with applied magnetic field gradients, to differentiate signals from various sections of the sample.

Nuclear Magnetic Moment the magnetic moment of a nucleus, usually given in nuclear magnetons.

Nuclear Spin the angular momentum of a nucleus expressed in units of $\hbar$.

Precession rotation of the axis of angular momentum of a nucleus about a magnetic field.

RF radio frequency.

RF Magnetic Field a magnetic field oscillating in the radio frequency (MHz) region.

Rotating Frame of Reference in NMR, the frame of reference rotating at the Larmor frequency about the applied magnetic field direction.

Spin Echo signal from the re-phasing of nuclear spins in the x - y plane, after the application of a 90° - 180° RF pulse sequence.

Spin-Lattice Relaxation (T_1) the return of the z nuclear magnetisation to its equilibrium value along the z axis.

Spin-Spin Relaxation (T_2) the return of the xy nuclear magnetisation to its equilibrium value of zero.

Static Magnetic Field a magnetic field which does not change with time.

References

- [1] *Pulsed/CW NMR Spectrometer (PS2-B)*. Instructor's Manual. TeachSpin, Inc.
- [2] Paul T Callaghan. *Principles of nuclear magnetic resonance microscopy*. Oxford science publications. Clarendon Press, Oxford [England], 1991.